**3** **(a)** **(i)** point which weight of ship M1

appears / seems to act A1

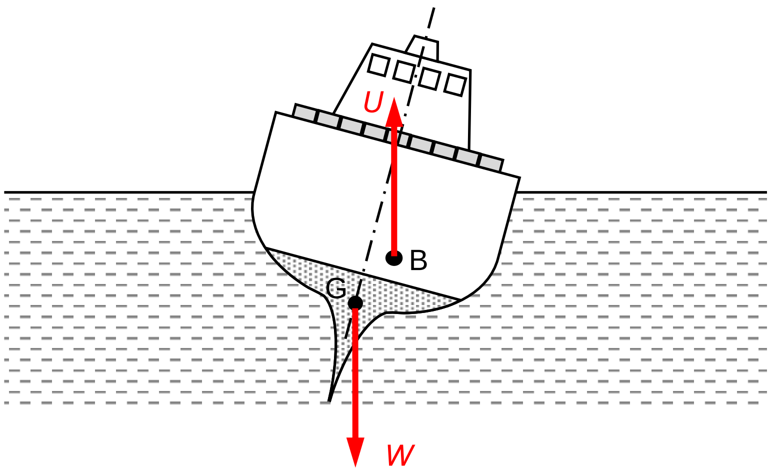
**(ii)** water pressure on lower surface greater than air pressure on upper surface of ship B1

**(b)** upthrust = weight

$1030\ V \times g = 2.20 \times 10^8 \times g$ C1

volume = $2.14 \times 10^5$ m$^3$ A1

**(c)** **(i)** vertically downward arrow for *W* at G, vertically upward arrow for *U* at B and length of *W* and *U* approximately equal B1

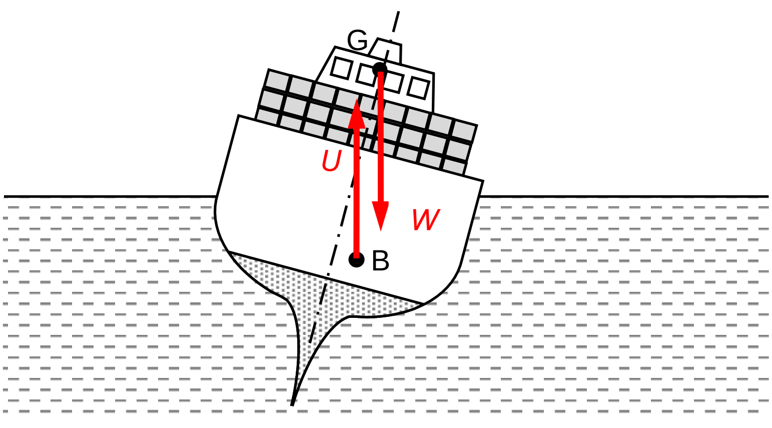



**(ii)** weight (or *W*) and upthrust (or *U*) forms a couple M1

which gives anticlockwise torque (or moment) A1

to right / correct the roll / tilt of the ship

**(d)** labelled weight through higher c.g. and labelled upthrust that produce overturning torque B1




weight and upthrust forms a couple that give clockwise torque (or moment) to overturn ship B1

**[Turn over**

**4** **(a)** **(i)** effective resistance across BN $= \left(\frac{1}{\ldots} + \frac{1}{\ldots}\right)^{-1} = R$